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Paper Code : PEC-CS702E/PEC-IT702F Cyber Security

UPID : 007558

Time Allotted : 3 Hours

Full Marks : 70

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer any ten of the following :

[1 x 10 = 10]

- (I) What is trojan horse?
- (II) Governments hired some highly skilled hackers for providing cyber security for the country or state. These types of hackers are termed as _____.
- (III) _____ is a code injecting method used for attacking the database of a system / website.
- (IV) Which of them is not a wireless attack?
 - a) Eavesdropping
 - b) MAC Spoofing
 - c) Wireless Hijacking
 - d) Phishing
- (V) Which of these is NOT a scanning type?
 - a. Null Scan b. Xmas Tree Scan c. SYN Stealth d. Cloud Scan
- (VI) Which malware does not clone or replicate itself through infection?
- (VII) Which is the least strong security encryption standard?
- (VIII) Which of the following is a component of cyber security?
 - a) Internet Of Things
 - b) AI
 - c) Database
 - d) Attacks
- (IX) Which of the following is not a step followed by cyber-criminals in data breaching?
 - a) Exfiltration b) Research and info-gathering c) Attack the system d) Fixing the bugs
- (X) Which of the following is defined as an attempt to harm, damage or cause threat to a system or network?
 - a) Digital crime.
 - b) Threats
 - c) System hijacking
 - d) Cyber Attack
- (XI) Which of the following is a type of cyber attack?
 - a) Phishing
 - b) SQL Injections
 - c) Password Attack
 - d) All of the above
- (XII) Define CIA Triad.

Group-B (Short Answer Type Question)

Answer any three of the following :

[5 x 3 = 15]

2. (i) Define Hacker. List Out the Types of hackers [5]
(ii) What is a Backdoor?
3. What is the difference between IDS and IPS? [5]
4. Explain port scanning. [5]
5. Discuss the significance of intellectual property rights (IPR) in cyberspace. [5]
6. Explain brute force attack and the ways to prevent it. [5]

Group-C (Long Answer Type Question)

Answer any three of the following :

[15 x 3 = 45]

7. How can be Intrusion Detection system is the backbone of Information system? Justify along with its categories? What do you mean by Application Security? [15]
8. With a neat diagram, explain the zachman framework for enterprise architecture. Discuss primitive models versus composite models. [15]
9. (a) Explain the major stages of a forensic investigation process. [10]
(b) Describe the role of auditing in cybersecurity and forensics. [5]
10. Explain the difference between Virus, Worms and Trojan Horse. What is the data security consideration? Explain in this reference Data backup security, Data archival security and Data disposal considerations. [15]
11. (a) Explain in detail the concept of social engineering. Describe its techniques and preventive measures. [10]
(b) What do you mean by SQL injection attacks. [5]

*** END OF PAPER ***